



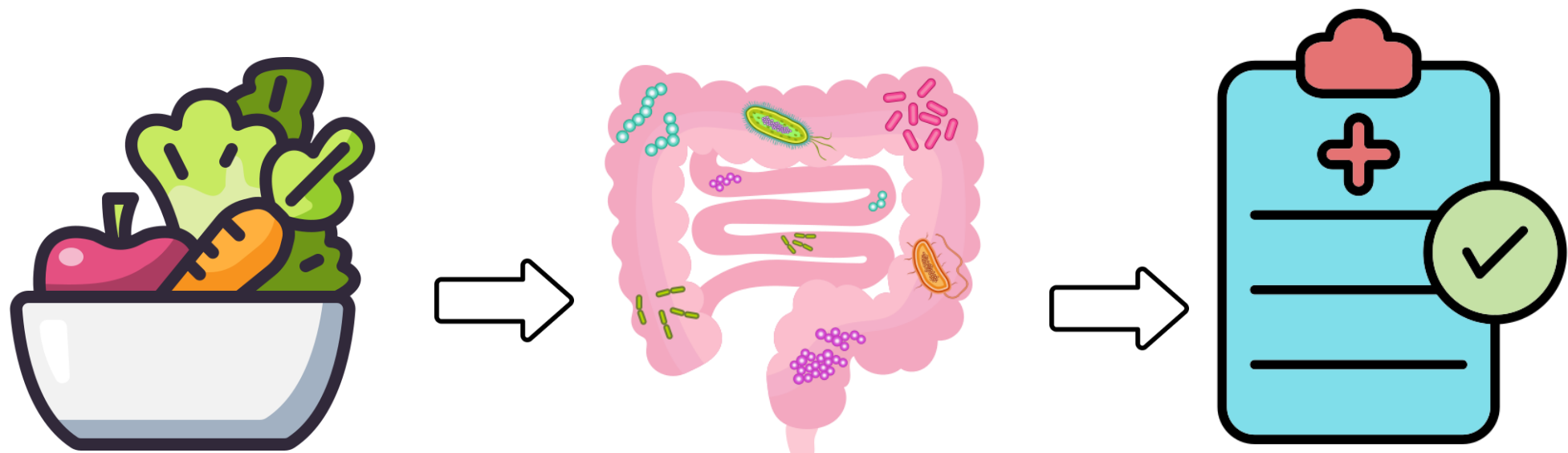
Dietary Intake Influences GI Health Outcomes via the Gut Microbiome and Metabolome in Ethiopian Schoolchildren



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Introduction

- Children in low- and middle-income countries are disproportionately affected by gastrointestinal (GI) symptoms¹, contributing to malnutrition, stunting, and reduced quality of life².
- In Ethiopia, young children have been found to have a high prevalence of GI symptoms including diarrhea and abdominal discomfort³.
- The human gut microbiome is a dense and complex ecosystem, populated by billions of microorganisms in the GI tract⁴, and it plays essential roles in digestion, nutrient uptake, and protection from pathogenic bacteria⁵.
- Diet is a major determinant of gut microbiome structure and metabolic activity⁶, yet the diet-driven molecular pathways linking microbial imbalances to GI symptoms remain poorly understood^{7,8}.



In this study, we evaluated how diet shapes gut microbiome and metabolome diversity, tested whether microbial or metabolic features mediate the relationship between diet and GI symptoms, and applied machine learning to predict bloating, diarrhea, abdominal pain, and low appetite using multi-omic features.

Methods

Study Setting and Design:

- School-based cross-sectional study (N = 57) in Hawassa City, Southern Ethiopia.

Measurements and Data Collection

- Interview-based questionnaire of parents/guardians; including Food Frequency Questionnaire (FFQ)⁴⁸ recalling the past three months, and demographic and health outcome measurements.
- Fecal sample collection for gut microbiome and metabolome sequencing and analysis.

Portion Size Calculation:

- Portion sizes were derived primarily from the FAO School Food Global Hub Ethiopian recommendations⁹, supplemented with the 2022 Ethiopia Food-Based Dietary Guidelines for foods not listed¹⁰.

Nutrient Calculation:

- Nutrient content (macro- and micronutrients) was calculated using the Ethiopian Food Composition Table Part IV¹¹, supplemented by NutriSurvey calculations¹².

Derived Dietary Variables:

- Food consumption score¹³, fruit/vegetable intake, injera intake, and animal product intake were calculated.

Microbiome & Metabolome α - & β -Diversity Analysis

- Alpha (Shannon¹⁴, Chao¹⁵) and beta diversity (Jaccard¹⁶, Bray-Curtis¹⁷) were calculated and tested against continuous and categorical variables, respectively. Analyses were adjusted for age, sex, and calorie intake.

Mediation Analysis

- Nutrients, microbes, and health outcomes were screened for mediation using linear and logistic regression models, adjusting for age, sex, and calorie intake. Mediation analyses were conducted in the *lavaan* R package¹⁸.

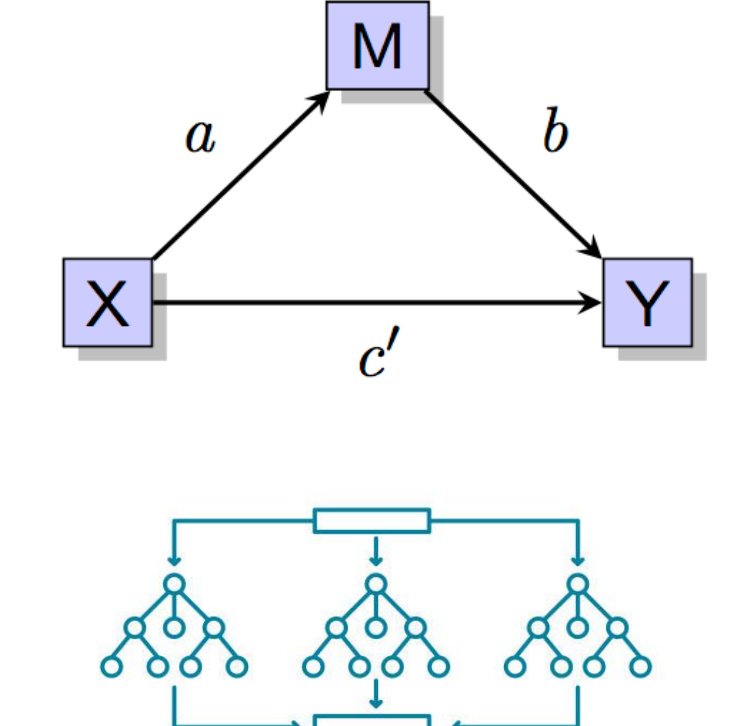
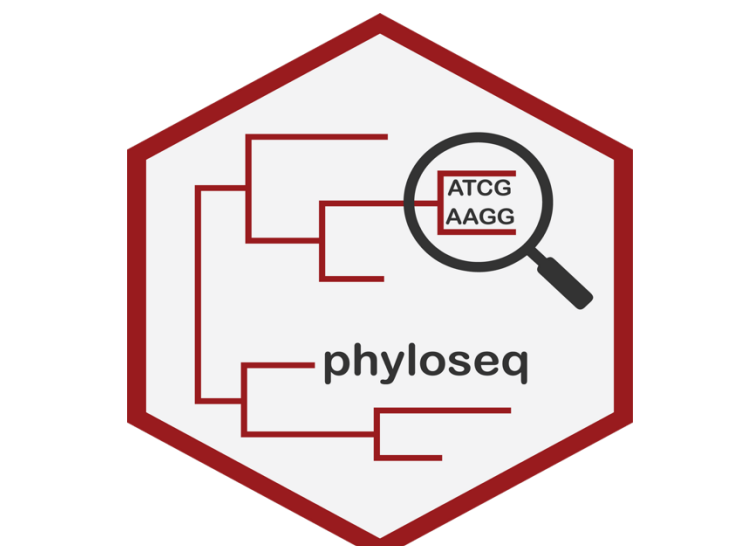
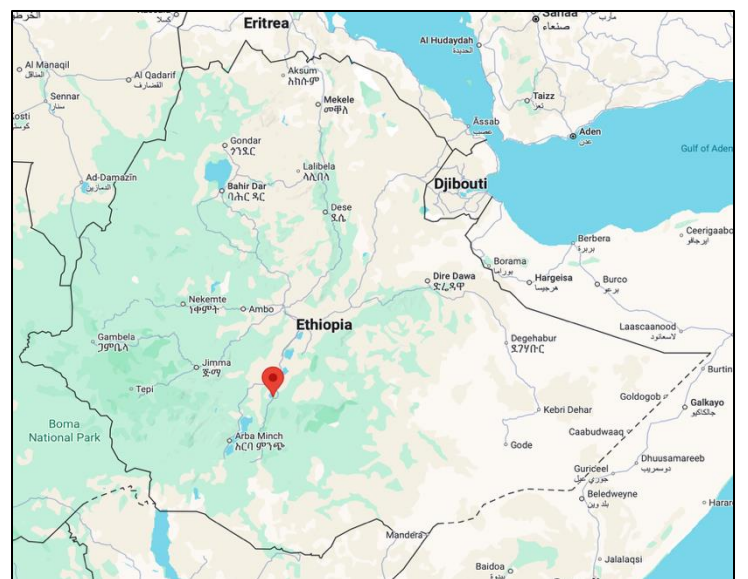
Machine-Learning Models

- XGBoost¹⁹ models predicting health outcomes were trained separately on diet, microbiome, and metabolomic features to identify top predictors. The top 20 features from each data type were then combined into a final integrated model.



Example of FFQ Question:

K1. Teff	1. Never or <1 * per month
	2. 1 * per month
	3. 2-3 * per month
	4. 1 * per week
	5. 2-3 * per week
	6. 4-6 * per week
	7. Every day
	8. More onetime per day



Results

Descriptive Characteristics of Study Population

Table 1. Descriptive Characteristics of the Study Population (N = 57)	
Characteristic	(Mean $\pm$ 2*SEM) or count(%)
Age (years)	10.8 $\pm$ 0.6
Weight (kg)	32.6 $\pm$ 2.4
Height (cm)	137.0 $\pm$ 3.8
Family size	4.9 $\pm$ 0.5
Male:Female	21(37%) : 36(63%)
Calories (kcal/day)	1,912.4 $\pm$ 147.8
Food Consumption Score	73.64 $\pm$ 3.55
Fruit/Vegetable (portions/day)	4.4 $\pm$ 0.4
% of children with 0-4 of GI symptoms	0: 26%, 1: 5%, 2: 17%, 3: 12%, 4: 39%

Table 1. Descriptive Characteristics of Study Population. Continuous variables are reported as mean $\pm$ 2*SEM; categorical variables are presented as counts.

Diet Shapes Microbiome Diversity (α - & β -diversity)

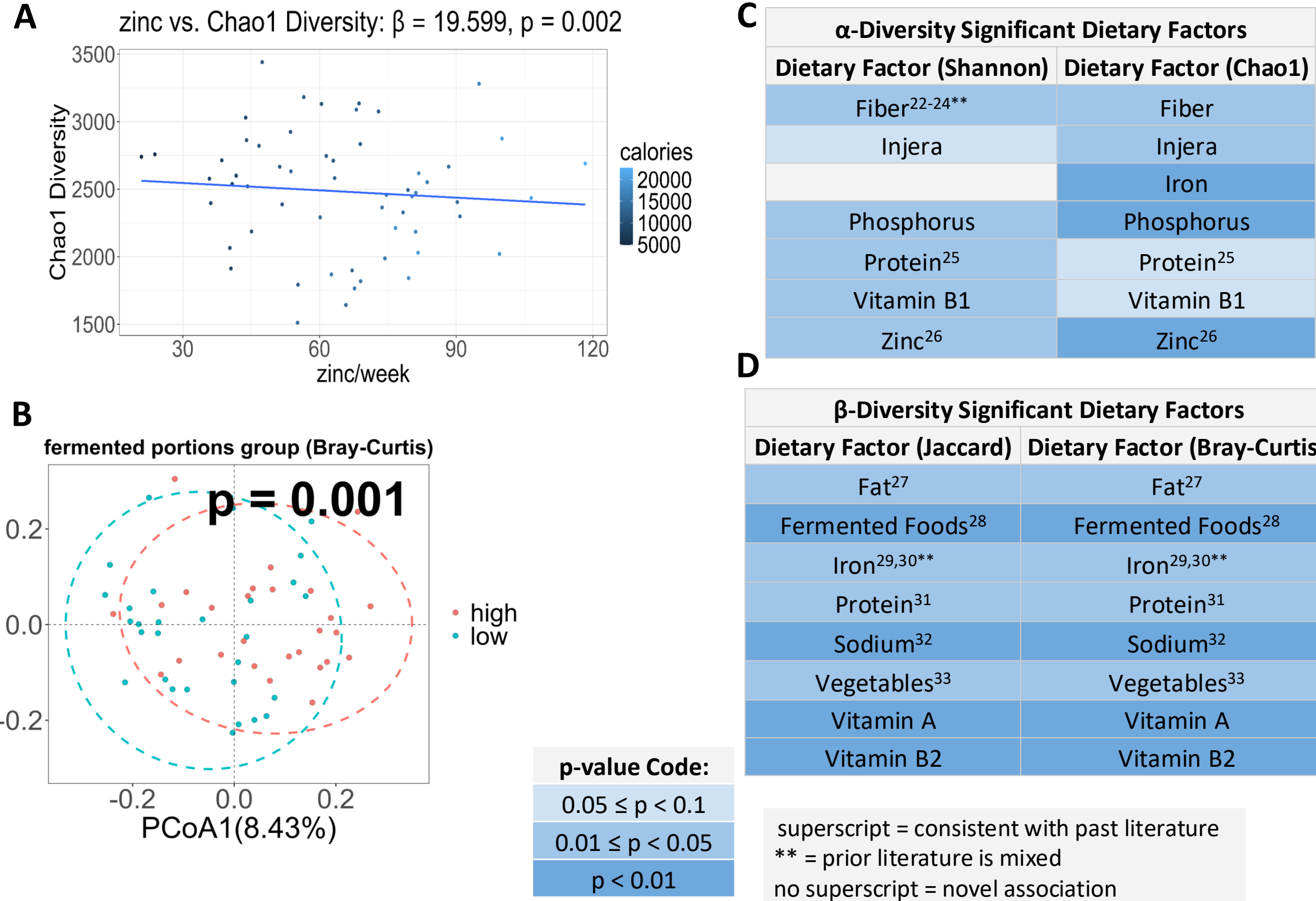


Figure 1. Diet shapes gut microbiome α - and β -diversity. (A) Higher zinc intake is associated with greater α -diversity (Chao1; $p = 0.002$), adjusted for calories, sex, and age. (B) Fermented food intake significantly alters β -diversity (Jaccard; $p = 0.001$), adjusted for calories, sex, and age. (C) Fiber, injera, iron, phosphorus, protein, vitamin B1, and zinc are significantly associated with α -diversity (Shannon, Chao1). (D) Fat, fermented foods, iron, protein, sodium, vegetables, vitamin A, and vitamin B2 are associated with shifts in β -diversity.

Diet Shapes Metabolome Diversity (α - & β -diversity)

A α -Diversity Significant Dietary Factor		B β -Diversity Significant Dietary Factors	
Dietary Factor (Shannon)	p-value Code:	Dietary Factor (Jaccard)	Dietary Factor (Bray-Curtis)
Calcium	0.05 $\leq$ p < 0.1	Carotene	Vitamin C
	0.01 $\leq$ p < 0.05		
	p < 0.01		

Figure 2. Diet shapes metabolome α - and β -diversity. (A) Calcium is significantly associated with α -diversity (Shannon). (B) Carotene (Jaccard) and Vitamin C (Jaccard and Bray-Curtis) are associated with shifts in β -diversity. All analyses are adjusted for calories, sex, and age.

Multi-omic Machine Learning Predicts GI Health Outcomes From Diet, Metabolome, and Microbiome

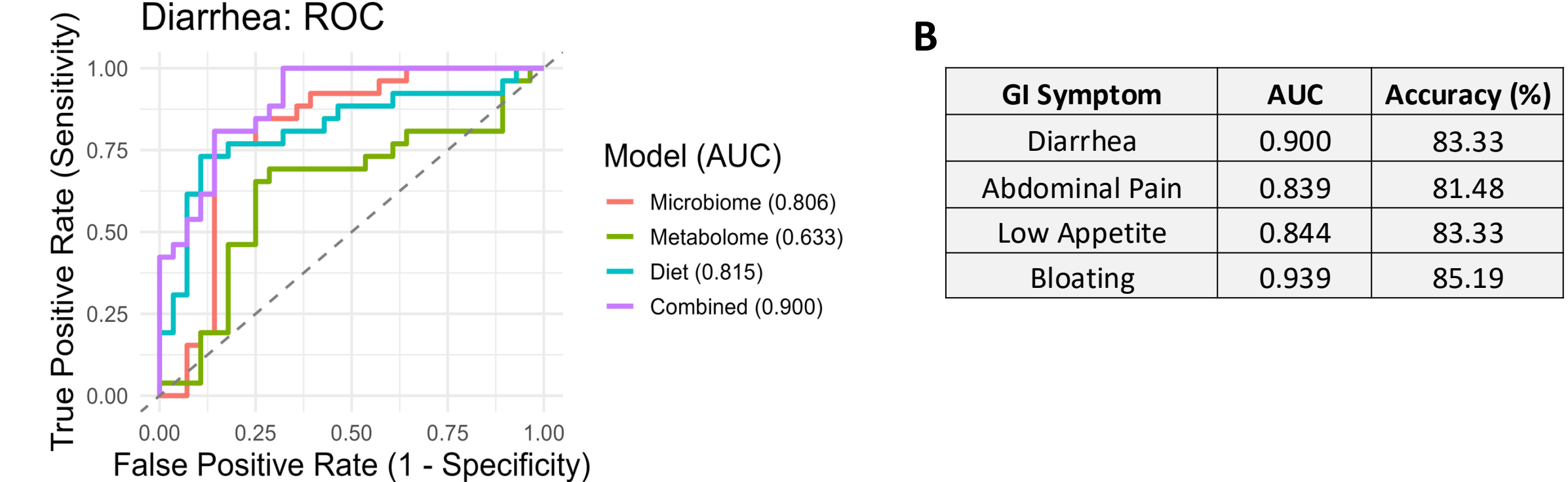


Figure 3. Machine learning prediction of GI symptoms. (A) ROC curves comparing diet, microbiome, metabolome, and combined models show that the combined model performs best (AUC = 0.900). Dietary factors and microbes strongly predicted diarrhea (AUC = 0.815) and (AUC=0.806), respectively, while the metabolome-only model performed poorly (AUC = 0.633). (B) Combined multi-omic models show strongest prediction for diarrhea (AUC = 0.900; 83.33% accuracy) and bloating (AUC = 0.939, 85.19 % accuracy) with strong performance for abdominal pain and low appetite.

Diarrhea : Top 10 Feature Importance

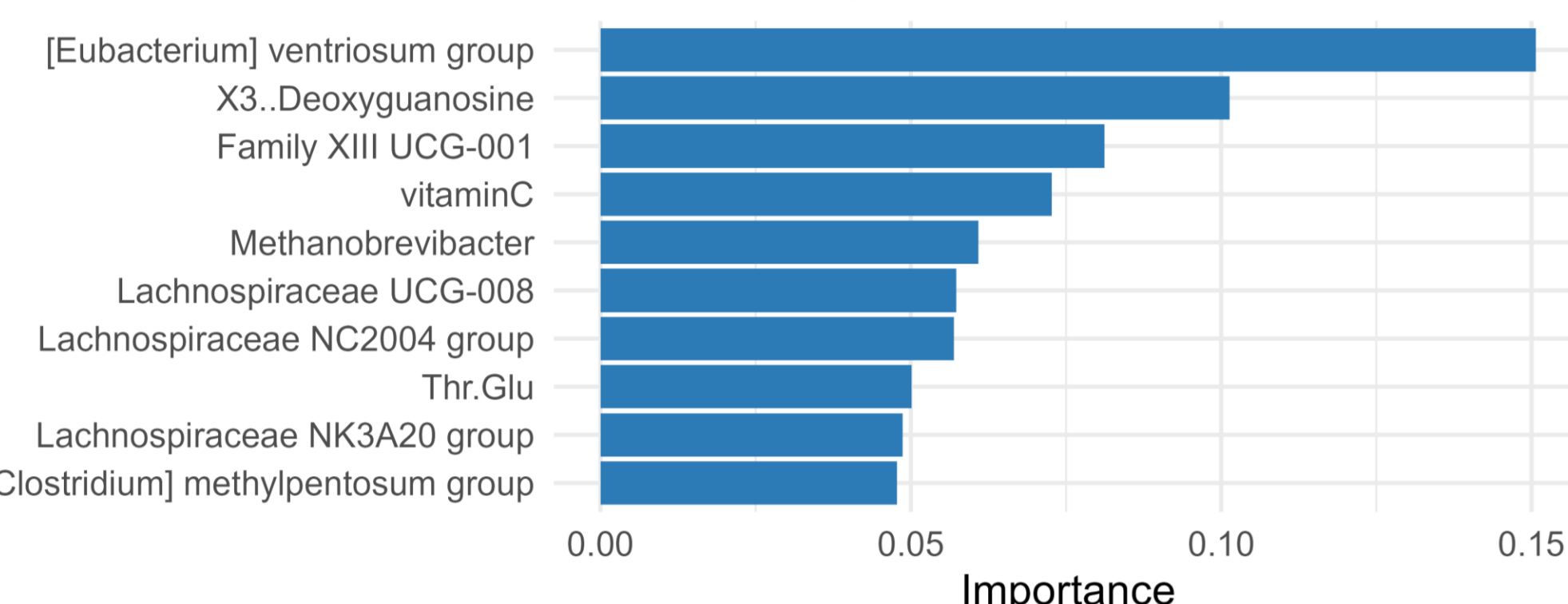


Figure 4. Top 10 feature importance for diarrhea prediction using XGBoost. Bar plot shows the 10 most important features in the diarrhea prediction model, including microbial taxa, metabolites, and vitamin C intake. Variables are ranked by their mean gain, which quantifies each feature's contribution to improving model accuracy across all boosted decision trees.

Microbes and Metabolites Mediate the Influence of Fruits and Vegetables on GI Health

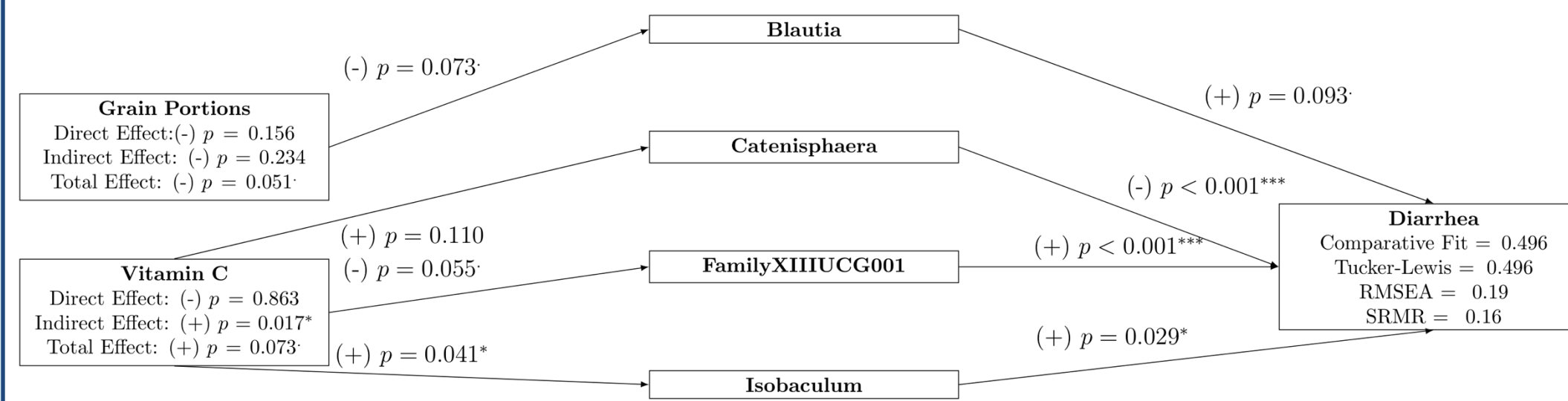


Figure 5. Dietary factors influence diarrhea through microbially mediated pathways. Grain portions showed a marginal total negative effect on diarrhea ($p=0.051$), and vitamin C showed a significant positive indirect effect ($p = 0.017$) with a non-significant direct effect indicating full mediation. Isobaculum showed the strongest mediation with significant associations from vitamin C to Isobaculum ($p = 0.041$) and from Isobaculum to diarrhea ($p = 0.029$). Model fit statistics are provided under the Diarrhea node.

Dietary Factor	→ (+/-)	Microbe / Metabolite	→ (+/-)	Health Outcome	Total Effect
Fruits + Vegetables	(+) ³⁴	<i>Hespellia</i>	(+)	Abdominal Pain	(+) ^{42,43}
	(+)	<i>Mobiluncus</i>	(+)		
	(+)	L-Threonine	(+)		
	(+)	X4-Hydroxy-4-methyl-2-pentanone	(+)		
	(+)	X2-Keto-3-deoxyoctonic-acid	(+)		
	(-)	<i>Atopostipes</i>	(-)		
	(-)	<i>Oceanicella</i>	(-)		
	(+) ³⁵	<i>Subdoligranulum</i>	(+) ³⁶		
Vitamin A	(-)	Ala-Val	(-)		
	(-)	Thr-Leu	(-)		
Vitamin C	(-)	<i>Papillibacter</i>	(-)		
	(-) ^{37*}	X3-4-Dihydroxybenzoic-acid	(+)		
	(-)	Hamamelitannin	(+)		
	(-)	X6-Oxo-octadecanoic-acid	(+)		
	(-) ^{38*}	<i>Blautia</i>	(+) ³⁹		
	(+)	Diisopropyl-phosphate	(+)		
	(-)	<i>Family-XII-UCG-001</i>	(+) ⁴⁰		
	(+)	<i>Isobaculum</i>	(+) ⁴¹		

superscript = consistent with past literature,** = prior literature is mixed, no superscript = novel association, * = inconsistent with past literature

Table 2. Dietary factors influence GI health outcomes through microbiome and metabolite mediated pathways. Each row shows a dietary factor, a significant microbial or metabolite mediator, and the GI symptom from mediation models. The first sign column gives the direction of the diet factor-mediator association and the second gives the direction of the mediator-symptom association ("+" = higher, "-" = lower). Only mediators with at least marginally significant ($p < 0.10$) indirect (mediated) effects are shown. The total effect column shows the overall direction of the nutrient-symptom association in our data.

Conclusions

- Diet strongly shapes gut microbial and metabolic diversity influencing health outcomes in schoolchildren in Southern Ethiopia.
- Multi-omic machine learning models accurately predicted diarrhea, abdominal pain, bloating, and low appetite, indicating that diet-shaped microbial and metabolic features offer relevant biological information towards understanding how and why GI symptoms occur.
- Fruits, vegetables, and vitamin C (commonly found in fruits and vegetables) were linked with several GI symptoms through specific microbes and metabolites, with some appearing as symptom promoting and others protective.

Future Directions

- This multi-omic approach should be replicated in much larger cohorts and across different regions of Ethiopia to improve generalizability and power.
- There should be further investigations into microbe/metabolite associations between fruits, vegetables, associated vitamins, and negative GI health outcomes in Ethiopia.

Acknowledgments

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References

